

What are the Odds?

Joint Probabilities for Continuous Random Variables

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Introduction

In May 2009, Patricia Demauro rolled two dice 154 consecutive times without rolling a 7, on her way to record winnings in the casino game craps. She surpassed the previous record of 118, a considerable margin of improvement. The odds of this occurrence were 1 in 1.56 trillion, or once in 1,560,000,000,000 (Keaton, 2021). An incredible feat, the probability of it happening almost defies understanding. Many events in the world can be represented with probabilities, common events and rare events alike. Quantifying these probabilities can provide us with a greater understanding and perhaps appreciation of the event itself, while also allowing us to make predictions for the future. This is the main utility of probability, its ability to project events which have not yet occurred. Thus, probability has applications in countless areas of life and countless fields of study. Almost everything which makes up the Earth is spawned from the probabilities involved in the development of life. From the probability of the exact conditions necessary for life to spring about in the first place to the probability of having blue eyes or red hair, it can all be represented mathematically.

The occurrence of two or more events simultaneously is inevitable due to the interconnectivity of the world. It is certain that everyone has pondered the chances of these simultaneous occurrences at some point. Perhaps it was the combination of specific occurrences in a sports game, or perhaps it was the occurrence of rain on the day of an exam. In fact, any two events which can be represented with probability could be hypothetically combined to create what is known as joint probability. How can joint probability be determined?

The answer, somewhat trivially, depends upon the chance of the first event occurring and on the chance of the second event occurring. This is complicated by the question of independence, which indicates whether either of the events have a direct effect on the other. While this can lead to complex and rigorous problems, this paper will focus on events which are independent of each other. That is, the occurrence of one event has no effect on whether the second event will occur, and vice versa. Determining the chance of simultaneous events will be examined in further detail.

Though it may not seem immediately applicable to everyday situations, the concepts of joint probability are perpetually a significant aspect of life. An adequate understanding of chance and independence allows for a greater understanding of the world, because representing a real-world occurrence mathematically strengthens its comprehension. A strong knowledge of joint probability will also improve one's ability to make predictions for the future.

Materials Needed

- **Pen or Pencil:** While simple joint probability questions may be possible by hand, it is always useful to write down the process. Colored pens or pencils can be even more useful
- **Paper (Graph):** A place to write down the work. Paper will aid in visualizing the problem. Graph paper is particularly helpful.
- **Formula Sheet (Optional):** Should have PDFs for common distributions. This will be very useful when the functions are *not* provided by the problem.

General Method

While there are several approaches to solving joint probability problems depending upon the nature of the question, the overarching method is as follows:

1. Finding and the Joint Probability Distribution (p. 3)
2. Finding the Bounds of Integration (p. 4)
3. Setting Up and Evaluating the Integral (p. 6)
4. Interpreting the Result (p.8)

Each of these steps has several sub-steps with corresponding intricacies, all of which will be discussed in the following sections. There are also several places in which mistakes are common, which will also be elaborated upon. The second and third steps are only necessary for continuous random variables.

The Problem

There is no limit to the number of possible problems in relation to joint probability, but this paper will focus on a problem of moderate complexity. The methods showcased can be applied to a wide range of problems.

A random variable X has the PDF $f(x; \theta) = \theta x^{\theta-1}$, $0 < x < 1$, and random variable Y has the PDF $f(y; \theta) = \theta y^{\theta-1}$, $0 < y < 1$. Find a general statement for the probability that $\frac{3}{4} \leq xy$.

Part 1: Finding the Joint Probability Distribution

Almost all joint probability problems involve random variables (see glossary). The most important thing to consider in this step is whether these variables, or events, are independent of each other. Many problems will provide this information, but others may require it to be proven. For the purposes of these instructions, we will assume that the two events are fully independent. Each of the steps will include accompanying examples.

WARNING: The following steps *only* apply when the two variables are independent!

0. Ensure that both random variables are continuous, and not discrete (see glossary).

Discrete random variables do not have a probability density *function* in the traditional sense, rather they have defined probabilities at specific values of the variable. They do not involve integration, and thus calculating their joint probabilities is simpler.

1. Find the Probability Density Function (PDF) for both variables.

$$f(x; \theta) = \theta x^{\theta-1}, \text{ and } f(y; \theta) = \theta y^{\theta-1}$$

2. Multiply the PDFs together.

$$f(x; \theta) \cdot f(y; \theta) = \theta x^{\theta-1} \cdot \theta y^{\theta-1} = \theta^2 x^{\theta-1} y^{\theta-1}$$

WARNING: Be careful in combining terms and in distributing, an error in simplifying the function will lead to inaccurate results.

3. Write down the result. This will be used in Parts 2 and 3.

$$f_{xy}(x, y; \theta) = \theta^2 x^{\theta-1} y^{\theta-1}$$

Part 2: Finding the Bounds of Integration

Recall that continuous random variables must be bounded by specific values because they can hypothetically equal any decimal value within their domain.

1. Write down the given conditions for the problem.

$$\frac{3}{4} \leq xy; 0 < x < 1, 0 < y < 1$$

2. Replace the inequality sign with an equals sign.

$$\frac{3}{4} = xy; 0 < x < 1, 0 < y < 1$$

3. Rearrange the equation so that it is a function of x or y .

$$y = \frac{3}{4x}; 0 < x < 1, 0 < y < 1$$

4. Sketch the function using its domain and range. Our example is bounded between 0 and 1 on both axes, so we can ignore the part of the function that lies in the third quadrant. The function is shown in Figure 1.

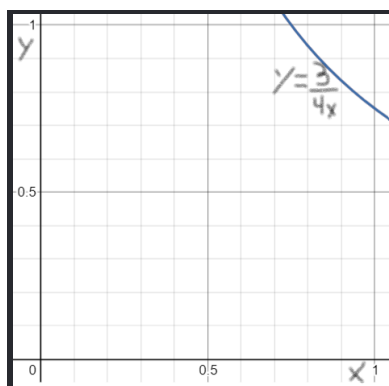


Figure 1.

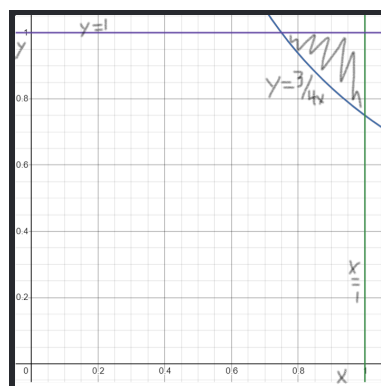


Figure 2.

5. Shade in the area defined by the problem. Since we want $\frac{3}{4} \leq xy$, this area will be above the curve from Figure 1. This is the area we want to calculate, represented by a jagged line in Figure 2.
6. Consider the values that y can assume within the shaded region and write them down. It can be anywhere between the function $y = \frac{3}{4x}$ to $y = 1$, shown by an arrow in Figure 3.

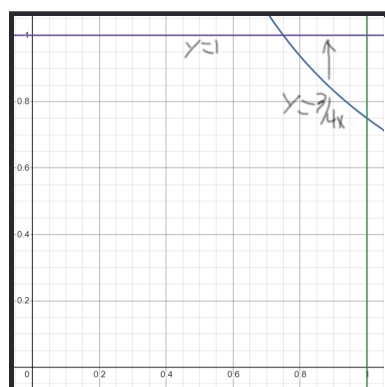


Figure 3.

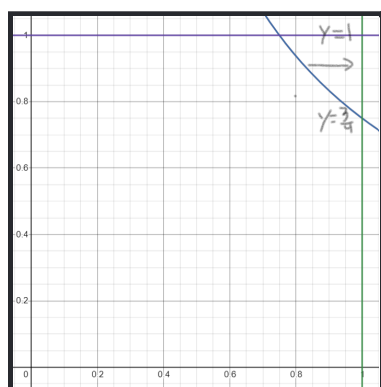


Figure 4.

7. Condense the graph to the right, as seen in Figure 4. This is shown with a horizontal arrow, and can be visualized as changing the two-dimensional section into a one-dimensional line.
8. Determine the values which x can assume by using the “condensed” graph

- a. Substitute 1 for x in the function $\frac{3}{4x}$. The result is $\frac{3}{4}$, which will be the lower bound because it is at the *bottom* of the line.
 - b. x cannot be any value greater than 1, 1 will be the upper bound for the x variable.
9. Write down both bounds, recalling which variable's bounds were determined first.
- y bounds (taken first): $\frac{3}{4x}$ to 1, x bounds (taken second): $\frac{3}{4}$ to 1.

Part 3: Setting Up and Evaluating the Integral

To calculate the shaded area, we must use the bounds in a double integral. Steps for this are as follows.

1. Write down the integral using the joint pdf and bounds.

$$\int_{\frac{3}{4}}^1 \int_{\frac{3}{4x}}^1 \theta^2 x^{\theta-1} y^{\theta-1} dy dx$$

2. Bring the constants to the front of the integral. A constant is a value entirely separate from both of the variables.

$$= \theta^2 \int_{\frac{3}{4}}^1 \int_{\frac{3}{4x}}^1 x^{\theta-1} y^{\theta-1} dy dx$$

3. Separate the “inside” integral from the “outside” integral. This is represented by brackets for the purpose of visualization

$$= \theta^2 \int_{\frac{3}{4}}^1 \left[\int_{\frac{3}{4x}}^1 x^{\theta-1} y^{\theta-1} dy \right] dx \rightarrow \int_{\frac{3}{4x}}^1 x^{\theta-1} y^{\theta-1} dy$$

WARNING: Do not forget about the outside integral! It will be brought back later.

4. Evaluate this inner integral.
 - a. Bringt x outside the integral. It is a constant since it is not yet being integrated.

$$x^{\theta-1} \int_{\frac{3}{4x}}^1 y^{\theta-1} dy$$

- b. Add one to the power of y, and divide by this new exponent.

$$= x^{\theta-1} \left[\frac{y^{\theta}}{\theta} \right], y = \frac{3}{4x} \text{ to } y = 1$$

WARNING: For the first integral, write the bounds as a function of the first variable which was integrated. This prevents the accidental combination or confusion of variables.

- c. Substitute the values for the bounds for y, subtracting the lower bound from the upper bound. Simplify using exponent rules.

$$= x^{\theta-1} \left(\frac{1^{\theta}}{\theta} - \frac{(3/4x)^{\theta}}{\theta} \right) = x^{\theta-1} \left(\frac{1}{\theta} - \frac{3^{\theta}}{\theta(4x)^{\theta}} \right)$$

5. Place the result within the “outside” integral.

$$\theta^2 \int_{\frac{3}{4}}^1 x^{\theta-1} \left(\frac{1}{\theta} - \frac{3^{\theta}}{\theta(4x)^{\theta}} \right) dx$$

6. Evaluate the integral.

- a. Distribute $x^{\theta-1}$ and simplify the expression using exponent rules.

$$= \theta^2 \int_{\frac{3}{4}}^1 \left(\frac{x^{\theta-1}}{\theta} - \frac{x^{\theta-1} 3^{\theta}}{\theta(4x)^{\theta}} \right) dx = \theta^2 \int_{\frac{3}{4}}^1 \left(\frac{x^{\theta-1}}{\theta} - \frac{3^{\theta}}{4^{\theta} \theta x} \right) dx$$

- b. Separate the integral into two parts using the sum rule.

$$= \theta^2 \left(\int_{\frac{3}{4}}^1 \frac{x^{\theta-1}}{\theta} - \int_{\frac{3}{4}}^1 \frac{3^{\theta}}{4^{\theta} \theta x} \right) dx$$

- c. Evaluate each integral.

$$\text{i. } \int_{\frac{3}{4}}^1 \frac{x^{\theta-1}}{\theta} = \left[\frac{x^{\theta}}{\theta^2} \right] x = \frac{3}{4} \text{ to } x = 1 \rightarrow \frac{1 - \left(\frac{3}{4}\right)^{\theta}}{\theta^2}$$

$$\text{ii. } \int_{\frac{3}{4}}^1 \frac{3^\theta}{4^\theta \theta x} = \left[-\frac{\ln(x) 3^\theta}{4^\theta \theta} \right] x = \frac{3}{4} \text{ to } x = 1 \rightarrow -\frac{\ln(\frac{3}{4}) 3^\theta}{4^\theta \theta} = -\frac{\ln(\frac{3}{4})(\frac{3}{4})^\theta}{\theta}$$

WARNING: For ii., Recall that $\ln(1)$ is defined to be 0, and $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$

- d. Combine these integrals and simplify the result by distributing and canceling terms (if possible).

$$\theta^2 \left[\frac{1 - \left(\frac{3}{4}\right)^\theta}{\theta^2} - \left(-\frac{\ln(\frac{3}{4})(\frac{3}{4})^\theta}{\theta} \right) \right] = \theta^2 \left[\frac{1 - \left(\frac{3}{4}\right)^\theta}{\theta^2} + \frac{\ln(\frac{3}{4})(\frac{3}{4})^\theta}{\theta} \right] = 1 - \left(\frac{3}{4}\right)^\theta + \theta \left(\frac{3}{4}\right)^\theta \ln\left(\frac{3}{4}\right)$$

7. Write down the final expression for the probability that $\frac{3}{4} \leq xy$.

$$P\left(\frac{3}{4} \leq xy\right) = 1 - \left(\frac{3}{4}\right)^\theta + \theta \left(\frac{3}{4}\right)^\theta \log\left(\frac{3}{4}\right)$$

Part 4: Interpreting the Result

The last step is to determine what the result means. In this problem, the parameter θ is unknown, hence the “generality” of the final expression. The expression for the probability would hold true so long as the value of θ does not cause the probability to be negative. That is, for any $\theta \geq 0$, the probability that x times y is greater than or equal to $\frac{3}{4}$ is defined by the function. While this is a relatively complex example, the methods can be applied to the joint probability of any two continuous random variables.

Simpler examples would involve discrete random variables, in which the only required steps are to find the precise probability of each event and multiply these values together, provided the events are independent. The result of this multiplication is the probability that both events occurred simultaneously. Joint probabilities can be complicated significantly if the two events are not independent. In these cases the covariance, a measure of the linear relationship between the variables, must be considered.

Glossary

Random Variable - A numerical expression of an event. Example: A random variable X is defined by the number of dots shown on a dice roll.

Discrete Random Variable - A random variable which is represented strictly with “counting” numbers. Example: A dice roll can take on values of 1, 2, 3, 4, 5, or 6.

Continuous Random Variable - A random variable which can be represented by any decimal value within a specific range of values. Examples could be height or weight.

Probability Distribution - Contains probabilities of occurrences for a random variable. Represented with a table, graph, or a function of the variable. Example: $X \sim N(\mu, \sigma^2)$, or random variable X follows a normal distribution with mean μ and variance σ^2 .

Probability Density Function (PDF) - Defines the probability function of a continuous random variable. Provides the probability that the variable will take on a value in a particular range.

Joint Probability Distribution - The combined distribution of two random variables. Can be represented with a table or a PDF.

Independence - The condition that the occurrence of one event has no effect on another, and vice versa. Example: what was had for breakfast has no effect on the weather.

Integral - A mathematical means of determining the area or volume, determined by values (bounds) which define it.

Domain - The values for which a function exists. e.g. the function $f(x) = x^2$ exists for all values of x , whereas the function $f(x) = \frac{1}{x}$ exists everywhere *except* $x = 0$, where it is undefined.

Works Cited

-Keaton, Brooke. (2021). The Amazing Story Of The Grandma And The Longest Craps Roll In History. *Casino.org*

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-Graphs and Charts created using <https://www.desmos.com/calculator>